Task 2: Process Model and Architecture

Final Project - Task 2 | Digital Loan Pre-Approval & Onboarding (MVP), Online Bank | Prepared by: [Your Name], Business Analyst | 09 November 2026

Part 1: To-Be BPMN Process Model

Scope: starts when a customer initiates a loan application and ends when the system issues an automated pre-approval decision, a decline, or routes the case to manual review (where an agent completes the decision).

Key workflow steps

#	Step	Lane	BPMN element
1	Start application and give consent	Customer	Start event + task
2	Capture personal, KYC and income information	Customer	User task
3	Completeness check	System	Exclusive gateway
4	Log consent	System	Service task (to Audit Log)
5	Verify identity (KYC + watchlist)	System	Service task
6	KYC result: pass / review / fail	System	Exclusive gateway
7	Perform soft credit pull (retry x3)	System	Service task
8	Evaluate credit decision	System	Exclusive gateway (business rules)
9	Route to auto-approve, auto-decline or manual review	System / Operations	Sequence flows
10	Agent reviews and decides	Operations	User task + gateway
11	Notify customer	System	Send task
12	Pre-approved or declined notice	Customer	End events

> **AI prompt used:** *Generate a BPMN model for an online bank's digital loan onboarding workflow. Include: application start, KYC data capture, soft credit pull, decision gateway for credit result, KYC failure path, manual review queue, automated approval, and customer notification. Use standard BPMN notation with three lanes (Customer, Onboarding System, Operations) and output Graphviz/Mermaid code. Every gateway must have labeled outgoing paths. Do not add steps that are not listed.*

To-Be BPMN diagram

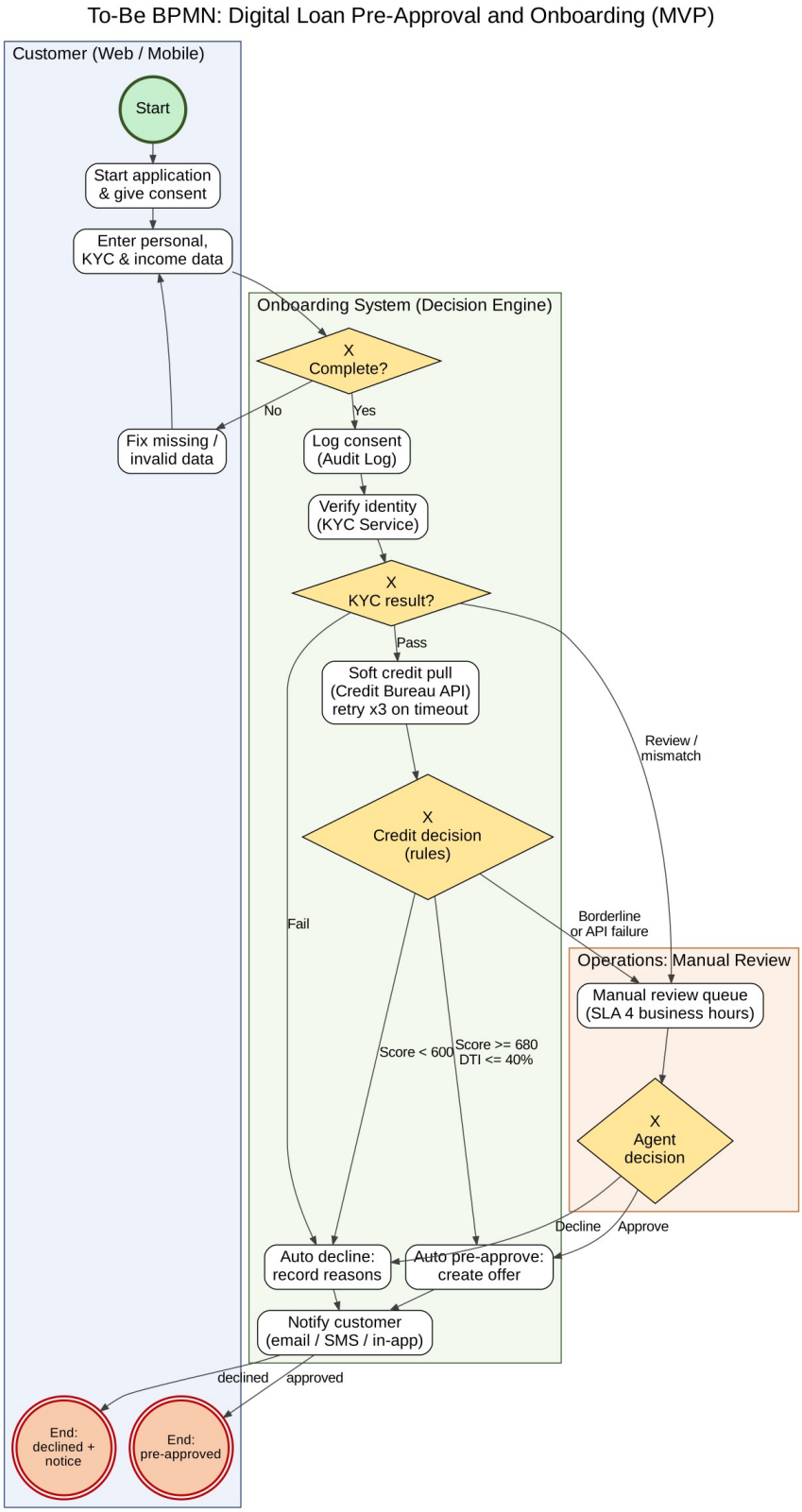


Figure 1. To-be BPMN with lanes Customer, Onboarding System and Operations (circle = start, double circle = end, rounded box = task, X diamond = exclusive gateway).

Validation check	Result
Decision gateway for credit score results	Yes: approve (score >= 680 and DTI <= 40%), decline (score < 600), manual review (borderline or API failure)
KYC verification branch	Yes: pass / review (partial match) / fail
Manual review lane	Yes: Operations lane with queue (4-hour SLA) and agent decision gateway
Customer notification end event	Yes: two end events, 'Pre-approved notice' and 'Declined / adverse action notice'
Incomplete data loop	Yes: 'Complete?' gateway loops back to data entry

Part 2: Customer Journey Map

> **AI prompt used:** *Create a customer journey map for a digital loan onboarding process with five stages: Discover > Apply > Pre-approve > Verify > Fund. Include customer actions, goals, emotions, and system touchpoints at each stage, plus pain points and improvement opportunities, for the persona 'busy professional applying on mobile'.*

Persona: Maya, 34, salaried professional, applies on her phone during a lunch break; wants a quick answer without hurting her credit score.

Stage	Customer actions	Goals	Emotions	Touchpoints	Pain points	Opportunities
Discover	Sees a personalized loan offer on web, app or email	Find a fair, fast loan	Curious, cautious about credit impact	Website, app banner, email	Unclear rates; fear of hurting credit score	Show 'checking won't affect your credit score' message and a rate range
Apply	Starts application, gives consent, enters personal and income details	Finish quickly on mobile	Hopeful, impatient	Web / mobile form, consent screen	Long forms; unclear why data is needed	Short form, progress bar, plain-language consent, save and resume
Pre-approve	Waits while KYC and soft pull run; sees decision	Instant answer	Anxious, then relieved or disappointed	Progress screen, decision page, email/SMS	Waiting with no feedback; vague decline	Decision in under 3 minutes; clear reasons and next steps
Verify	Confirms identity; may be contacted if routed to review	Get verified without hassle	Frustrated if asked to repeat info	KYC check, agent contact, notifications	Not knowing review status	Status tracker with expected time (4-hour SLA)
Fund	Reviews final offer, signs, receives money	Get funds fast	Satisfied, relieved	E-sign, account, notification	Delays between approval and funding	Clear funding timeline (full funding automation out of MVP scope)

Emotional curve: cautious (Discover) > impatient (Apply) > anxious then relieved (Pre-approve) > frustrated if delayed (Verify) > satisfied (Fund). Biggest risk to conversion: Apply and Verify.

Part 3: Component and Data Flow Diagram

> **AI prompt used:** *Generate a system component and data flow diagram for a digital loan onboarding process. Include: Web app, Mobile app, API Gateway, Credit Bureau API, KYC Service, Core Banking database, and Audit Logging. Show direction of data flows and label each flow with the data it carries.*

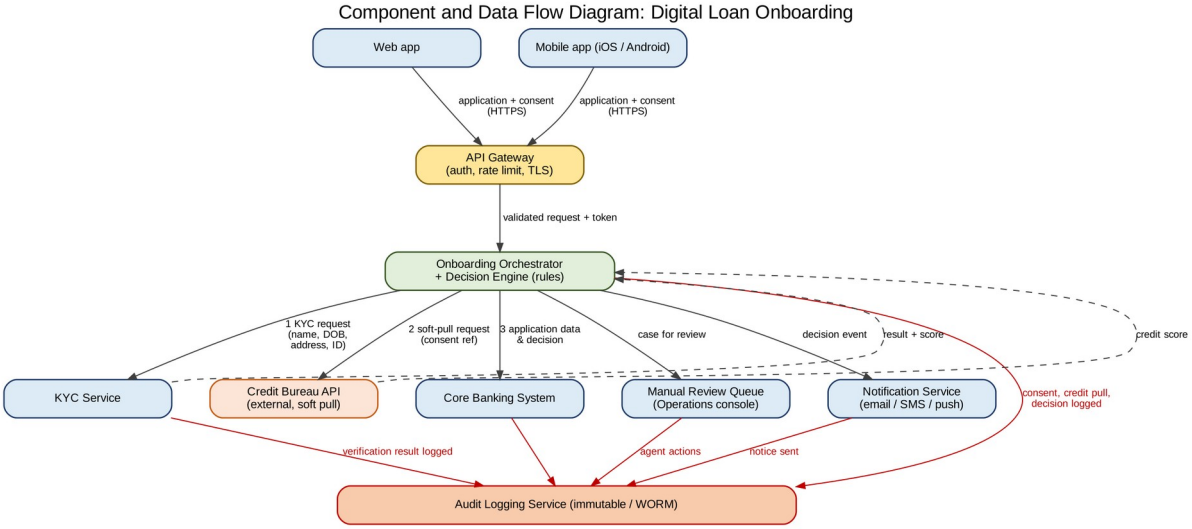


Figure 2. Components and data flows (solid = request, dashed = response, red = audit logging).

#	From > To	Data	Notes
1	Web / Mobile > API Gateway	Application data, consent	HTTPS, OAuth token, rate limiting
2	API Gateway > Orchestrator	Validated request	Schema validation, auth
3	Orchestrator > KYC Service > Orchestrator	KYC request (name, DOB, address, ID) > verification result + match score	PII minimized; result stored in KYCCheck
4	Orchestrator > Credit Bureau > Orchestrator	Soft-pull request with consent reference > credit score	External; retry 3x with backoff
5	Orchestrator > Core Banking	Application record and decision	Returns application record ID
6	Orchestrator > Manual Review Queue	Case with trigger reason	Agent decision returned
7	Orchestrator > Notification Service > Customer	Decision event > email/SMS/push	No sensitive data in SMS
8	All services > Audit Logging	Consent, data access, decisions, agent actions	Append-only, tamper-evident